

3D MODEL · SPEC CONVERSION

E3D Macro File(PIPE) ---> PCF(CAESAR II)

Converts an E3D macro file (PIPE) into CAESAR II PCF input. Tkinter GUI.

Key Purpose

Why this program

1. Turn an E3D piping macro straight into CAESAR II input

– Parses the .MAC text and the catalogue offline, then writes ready-to-import PCF files

→ **Effect:** The routed line reaches the stress model in minutes with no manual re-keying and no transcription error

2. No E3D login, session or console required

– Reads local macro and catalogue files only — pick pipes, queue them, export

→ **Effect:** Any designer can produce PCF anywhere, with no live E3D seat tied up and no license waiting

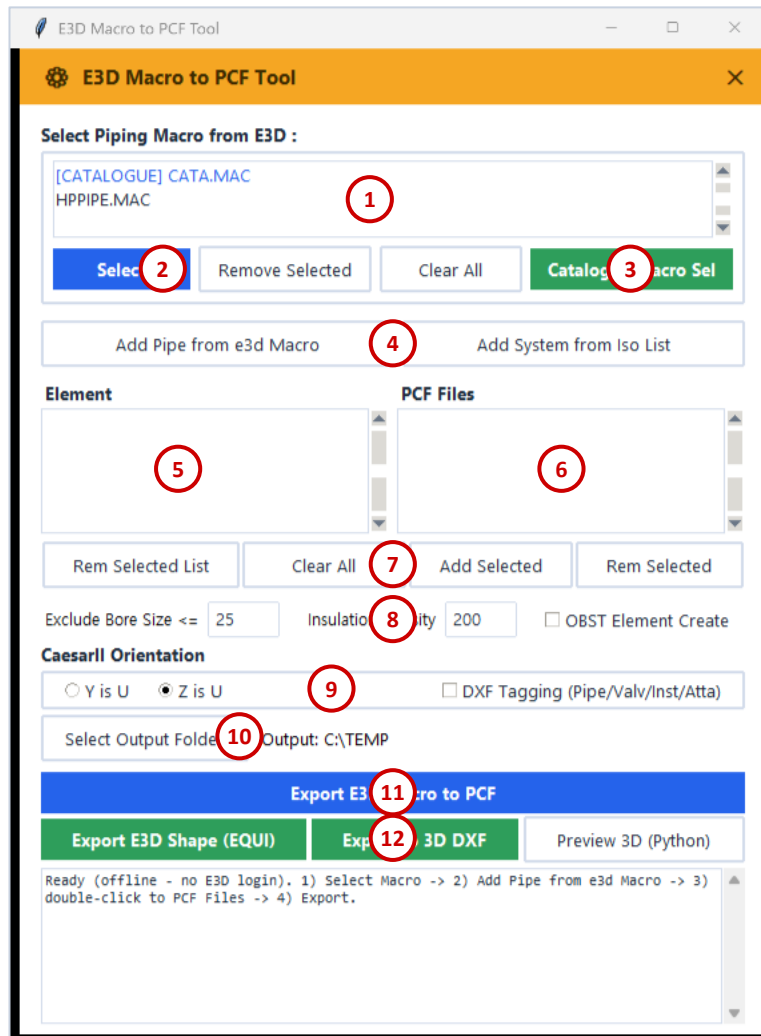
3. Reuse the same model as 3D shape, DXF or a quick preview

– Catalogue geometry rebuilds EQUI primitives, a 3D DXF mesh, or a Python 3D view

→ **Effect:** The line can be checked visually and re-modelled for clash review without redrawing it by hand

Required inputs : E3D piping macro (.MAC) + catalogue macro (CATA.MAC)

Main Window – Item Guide



1. Selected piping macros (+ tagged catalogue)
2. Select / Remove / Clear macro files
3. Catalogue Macro Select (CATA.MAC)
4. Add Pipe from macro / Add System from ISO list
5. Element list — parsed PIPE names
6. PCF Files list — export queue (double-click to add)
7. List actions — add / remove between lists
8. Exclude bore, insulation density, OBST option
9. CAESAR II orientation (Y-up / Z-up) + DXF tagging
10. Select output folder + current output path
11. Export E3D Macro to PCF
12. Export EQUI shape / 3D DXF / Preview 3D

Overview

What it does

1. Converts an E3D macro file (PIPE) into CAESAR II PCF input
2. Tkinter GUI

Category: 3D Model · Spec Conversion

Folder: CAESARII/E3D_MacroToPCF

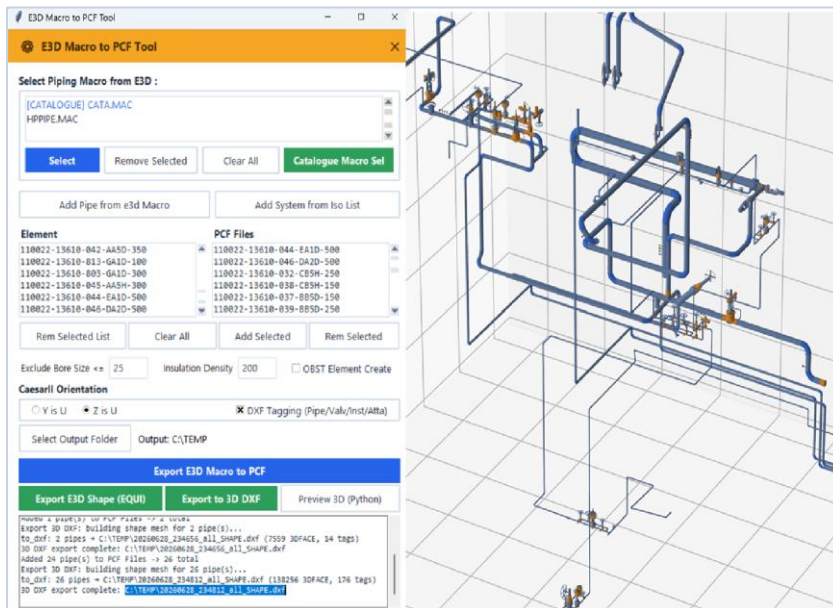
Main scripts: macrotopcf_gui.py

Scripts / Status: 7 scripts · Operational

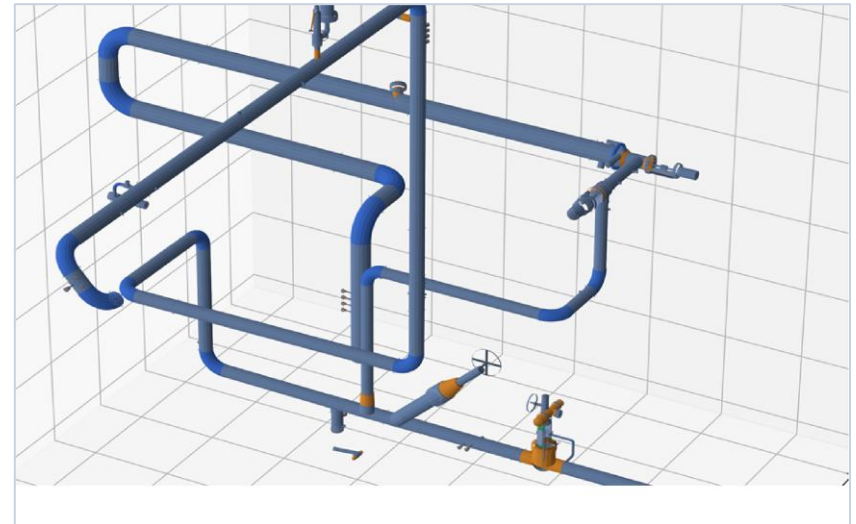
Tags: E3D · Macro · PCF · CAESAR II

Output Samples

Output Samples (1/5)



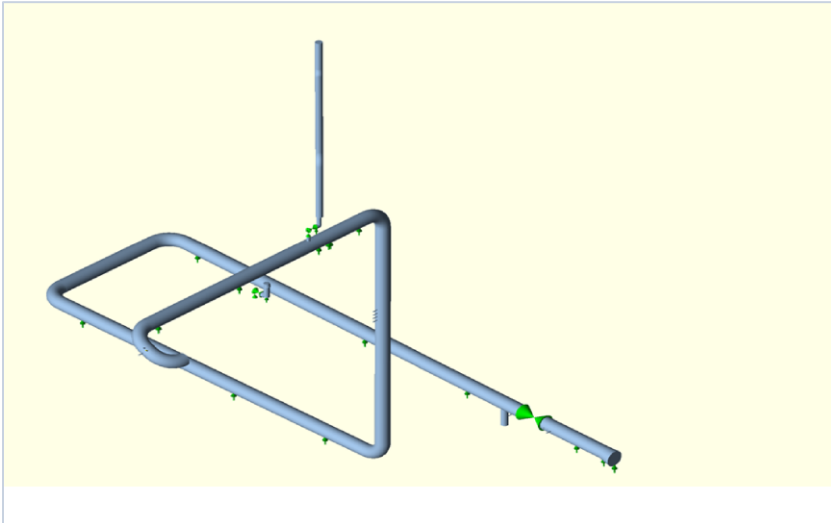
E3D Macro to PCF - Input



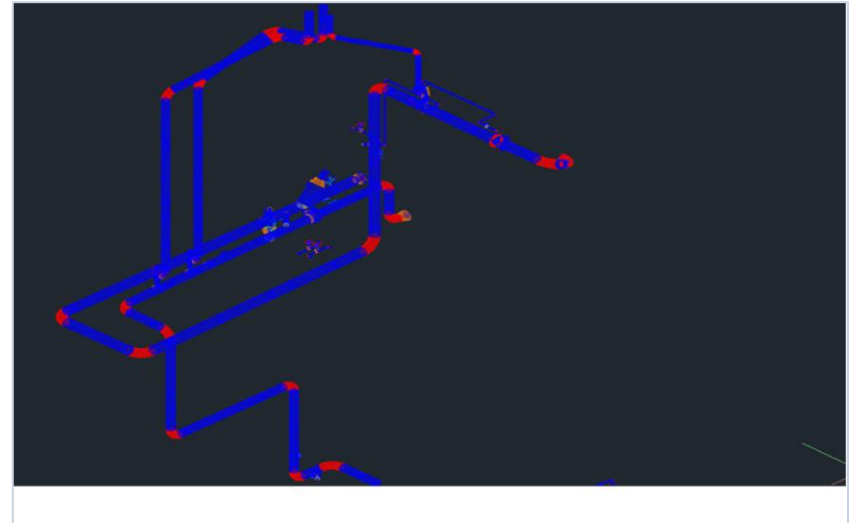
E3D Macro to PCF - Input 2

Output Samples

Output Samples (2/5)



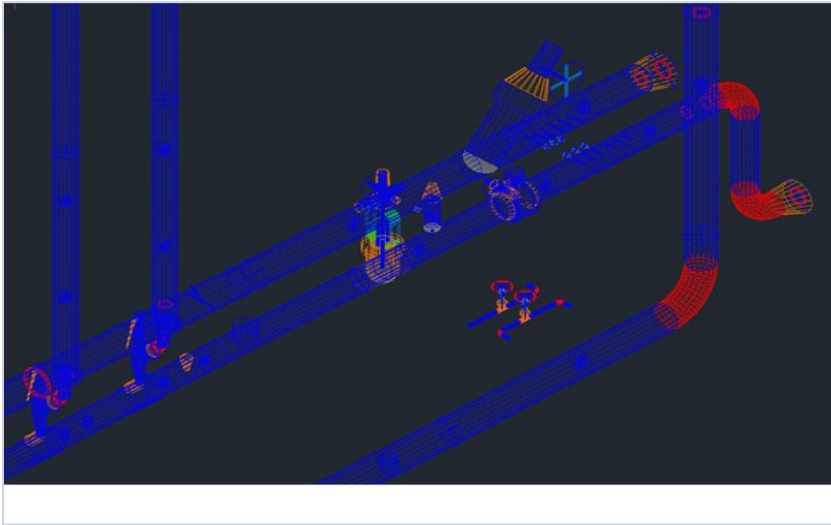
E3D Macro to PCF - Output



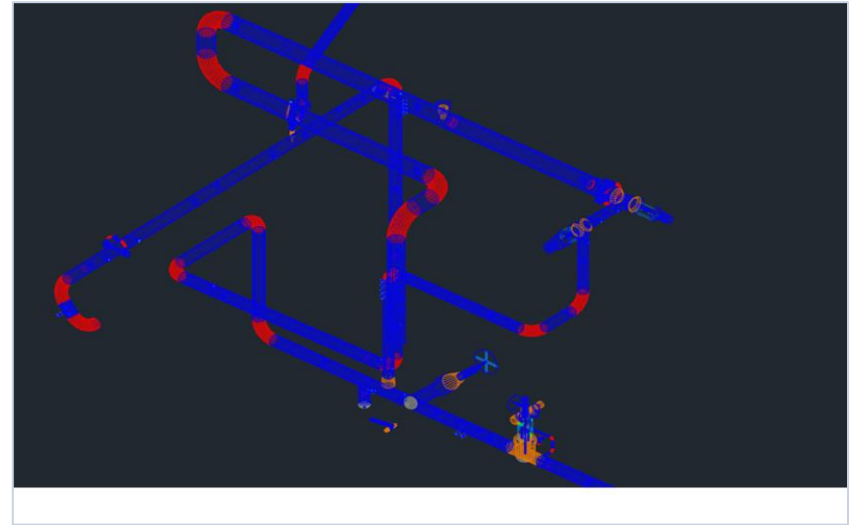
E3D Macro to PCF - DXF Output 1

Output Samples

Output Samples (3/5)



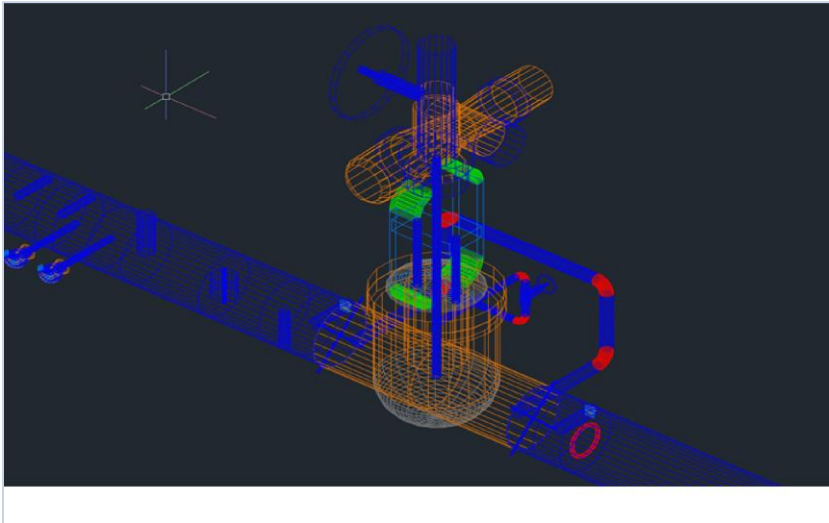
E3D Macro to PCF - DXF Output 2



E3D Macro to PCF - DXF Output 3

Output Samples

Output Samples (4/5)



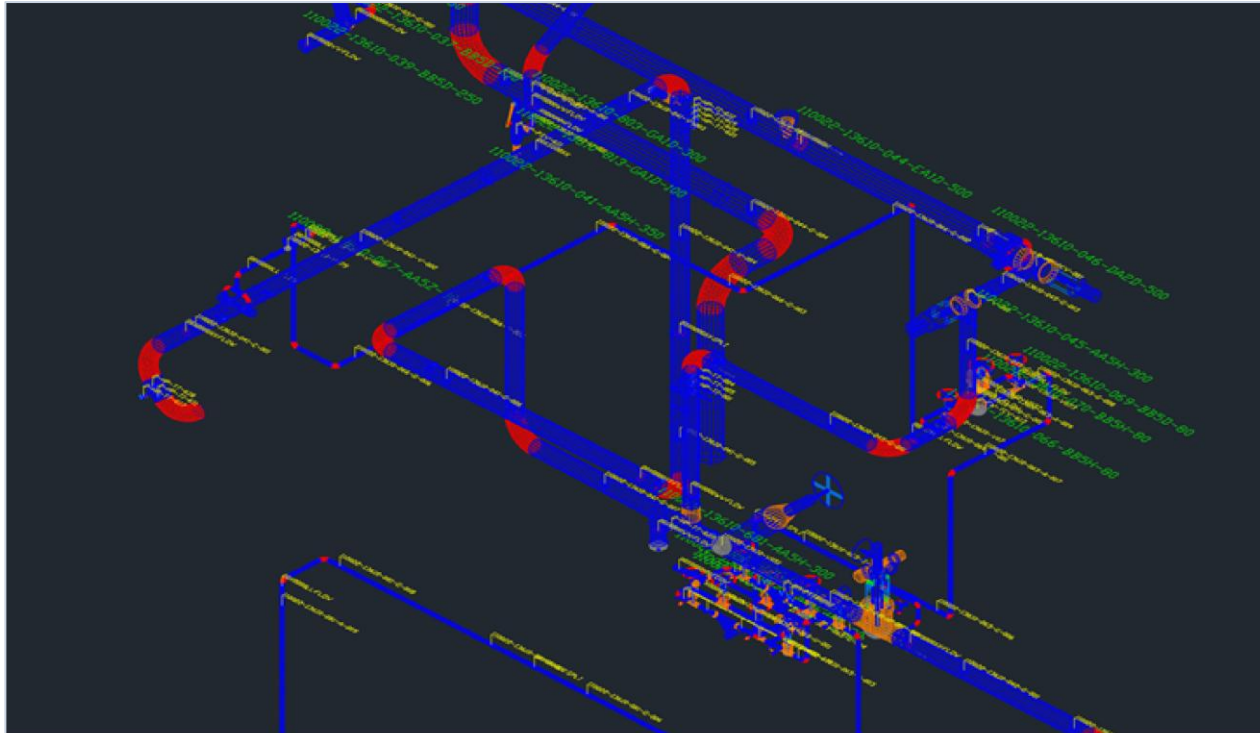
E3D Macro to PCF - DXF Output 4



E3D Macro to PCF - DXF Output 5

Output Samples

Output Samples (5/5)



E3D Macro to PCF - DXF Output 6

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: 3D Model · Spec Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: macrotopcf_gui.py.
- Output samples and the program window above reflect the current build.

Thank You

E3D Macro File(PIPE) ---> PCF(CAESAR II) · Plant Engineering Total Service System